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Using Community Engagement to Develop a Web-Based Intervention for Latinos about the HPV Vaccine

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Human papillomavirus (HPV) infection is pervasive among sexually active women and men, and Hispanic women are at particularly high risk as they have higher rates of invasive cervical cancer compared to other racial or ethnic groups in the United States. There is a need for interventions to increase HPV vaccination among this high-risk population. This study investigated how to modify a previously developed web-based intervention that provided individually tailored information about HPV to improve its use among the Latino population. A community-oriented modification approach incorporated feedback from a community advisory committee, and focus groups among the Latino population, to modify the intervention. Several themes emerged including a need for basic information about HPV and HPV vaccination, changes to make the intervention appear less clinical, and incorporation of information addressing barriers specific to the Latino community. This work was done in preparation for a randomized trial to assess the impact of this modified intervention on HPV vaccination attitudes and uptake among Latino young adults and parents of adolescents. If effective, our intervention could be a resource for reducing HPV vaccination concerns, improving immunization rates, and educating Latinos about HPV and the HPV vaccine outside of the time boundaries of the traditional clinical encounter.

Human papillomavirus (HPV) infection is the most common sexually transmitted disease worldwide, affecting more than 6 million individuals annually in the United States alone (Centers for Disease Control and Prevention, 2004; Markowitz et al., 2014). HPV infection causes virtually all cases of cervical cancer and abnormal Pap smears, a high proportion of vaginal and vulvar cancers in women, penile cancers in men, and most cases of genital warts and anal cancers in both genders. Overall, ~26,900 HPV-related cancers per year are diagnosed, with cervical cancer contributing ~ 12,000 cases annually (Centers for Disease Control and Prevention, 2012; Gargano et al., 2012; Hopenhayn et al., 2014; SEER; Steinau et al., 2013).

While HPV infection is nearly ubiquitous among the sexually active population (Markowitz et al., 2014), the morbidity and mortality from HPV-related diseases disproportionately affect minorities (National Cancer Institute). Hispanic women are at particularly high risk as they have the highest rates of invasive cervical cancer when compared to all other racial or ethnic groups in the United States (Cokkinides, Bandi, Siegel, & Jemal, 2012; National Cancer Institute; Siegel et al., 2015).

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The 2006 licensure of effective HPV vaccines ignited hope that the long-standing disparities in HPV-related diseases might be mitigated (National Cancer Institute). Although Hispanic adolescents currently have higher rates of HPV vaccination than other populations (Reagan-Steiner et al., 2015), coverage is still far below national goals (U.S. Department of Health and Human Services, 2010) with only 46.9% of adolescent Hispanic females having completed the three-dose vaccine series as of 2014 (Reagan-Steiner et al., 2015) and only 30% of young adult Latinas having received even one dose as of 2013 (Williams et al., 2015). These data underscore the critical need for interventions to increase HPV vaccination among this high-risk population.

Concerns about the safety and efficacy of HPV vaccination have been identified as major barriers to vaccine uptake among both adolescents and young adults (Rambout, Tashkandi, Hopkins, & Tricco, 2014). These barriers often cannot be adequately addressed during a typical clinical encounter due to time constraints (Davis et al., 2001; Gowda, Schaffer, Dombkowski, & Dempsey, 2012a; Kempe et al., 2011; O'Leary et al., 2012). Finding ways to address concerns outside of the clinician–patient encounter, such as through web-based applications, may help circumvent this problem (Glasgow, Bull, Piette, & Steiner, 2004; Jimbo, Nease, Ruffin, & Rana, 2006). While internet-delivered interventions to provide health prevention information outside of the clinical setting have been developed for many domains, there is a paucity of research on using such methods to deliver HPV vaccine-related information (Laugesen,

Hassanein, & Yuan, 2015; Springvloet, Lechner, de Vries, Candel, & Oenema, 2015; Ubhi, Michie, Kotz, Wong, & West, 2015; Yardley, Morrison, Bradbury, & Muller, 2015). Prior research suggests that such a modality for conveying information about HPV vaccines would be acceptable to parents (Gowda, Schaffer, Dombkowski, & Dempsey, 2012b).

Tailoring information to each individual's characteristics, preferences, and values has proven effective for improving compliance with several preventive health behaviors, including vaccination (Dempsey, Schaffer, Barr, Ruffin, & Carlos, 2011; Gerend, Shepherd, & Lustria, 2013; Gowda, Schaffer, Kopec, Markel, & Dempsey, 2012; Kreuter, Caburnay, Chen, & Donlin, 2004; Kreuter, Strecher, & Glassman, 1999; Kreuter & Wray, 2003). We previously developed a web-based intervention, called *Teen VaxScene*, which provided individually tailored information about HPV (and other adolescent vaccines) to parents (Dempsey, Maertens, Beaty, & O'Leary, 2015). A version of this intervention was demonstrated to effectively improve HPV vaccine acceptability, but was evaluated among primarily Caucasian parents (Dempsey et al., 2011). We hypothesized that the HPV component of *Teen VaxScene* would need to be modified to improve the intervention's use among the Latino population specifically.

We considered several modifications to improve the technical aspects of the intervention, including user-friendly navigation, page design, and inclusion of interactive elements, (Ludden, van Rompay, Kelders, & van Gemert-Pijnen, 2015; Schneider, van Osch, & de Vries, 2012; Shoup et al., 2015), along with ensuring readability (Rowsell et al., 2015; Schneider et al., 2012). However, we hypothesized that most modifications should focus on "culturally tailoring" the intervention for the Latino population. Prior research has identified several culturally specific beliefs about HPV vaccination among the Latino population broadly and amongst subsets within it (DiClemente, Murray, Graham, & Still, 2015; Glenn et al., 2015; Pierre Joseph et al., 2014; Podolsky, Cremer, Atrio, Hochman, & Arslan, 2009; Watts et al., 2009; Yeganeh, Curtis, & Kuo, 2010). Cultural tailoring, which recognizes each group's values, beliefs, and behaviors to provide a relatable context and meaning to the information provided (Morrison, Moss-Morris, Michie, & Yardley, 2014) would likely be important for effectively conveying information about HPV vaccines within our intervention. Few prior studies have tested the effect of cultural tailoring of vaccine-related interventions (DiClemente et al., 2015). Therefore, the goal of this study was to gather an in-depth understanding of how to modify the HPV component of the *Teen VaxScene* intervention to improve its use in general, as well as for the Latino population specifically. This work was done in preparation for a large randomized trial that would assess the impact of this newly modified intervention on HPV vaccination attitudes and uptake among Latino parents of adolescents and young adults.

Methods

Study Overview

Our original intervention, *Teen VaxScene*, created customized information about adolescent-targeted vaccines through an interactive interface and has been described elsewhere (Dempsey et al., 2015). Briefly, using the intervention requires participants to complete a baseline survey after which they are directed to webpages that are customized using a "tailoring engine" (University of Michigan Center for Health Communications Research) to generate content using the survey responses. Customization includes: tailoring of photographs to match participants' race and gender; prioritizing information about vaccines that matches participants' specific concerns, beliefs, or experiences; referencing demographic characteristics such as first name, age, and gender throughout the webpages' text; and using reflective statements to reiterate parents' stated opinions.

In the current study, we used a systematic, iterative, community-oriented modification approach to modify *Teen VaxScene* for the Latino population. This approach incorporated feedback from a community advisory committee (CAC) that represented our target population and data from focus groups among members of the target population. All focus groups and CAC meetings occurred over a 12-month period. Study procedures were approved by the Colorado Multiple Institutional Review Board.

Community Advisory Committee

Our CAC was derived using a purposeful snowball recruitment strategy and was comprised of five Latina parents of female adolescents and four young adult Latinas. Participants were recruited from organizations that serve Hispanic families, staff from the clinics where the planned randomized trial was to take place, and the general community. The first CAC meeting focused on ways to modify *Teen Vaxscene* for Latinos. These suggestions were used to develop materials for focus groups with English- and Spanish-speaking young adults and parents of adolescents. At three subsequent meetings, the CAC helped make iterative revisions to the logo, color scheme, and content that incorporated their evolving opinions about the intervention. The CAC also assisted in interpreting results from the focus groups and operationalizing these results into intervention modifications. CAC feedback was collected by two members of the study team through detailed written meeting notes and best practices documents, and then combined to complete a final summary. Notes were shared with CAC members to ensure that relevant information was adequately captured.

Focus Groups

Focus groups provided broader input about modifying *Teen Vaxscene* content and design for the Latino population, and to understand the perspectives of this population with regard to factors that impact decisions around initiating and completing the recommended HPV vaccine series. Participants were recruited from five clinics that serve low-income, primarily Latino, medically underserved, and migrant farmworker populations as priority clientele, and from the general Latino community. Recruitment occurred via advertisements posted in clinical sites and community venues. Eligible *parents* self-identified as being Hispanic or Latino, were ≥18 years old, were able to converse in

Table 1. Composition of focus groups.

Focus Groups	English (*N*)	Spanish (*N*)
By gender		
Male	3	0
Female	21	23
By group		
Parents (with daughters 11–17 years old)	10	17
Young adults (18–26 years old)	14	6

English or Spanish, and reported having an adolescent between the ages of 9 and 17 years who had not received all three doses of the HPV vaccine. Eligible *young adults* self-identified as being Hispanic or Latino, were between 18 and 26 years old, were able to converse in English or Spanish, and had not received all three doses of the HPV vaccine.

Six 90-minute focus groups with a total of 20 Latina young adults aged 18–26 and 27 Latino parents of adolescents between the ages of 9 and 17 were conducted by trained members of the research team (Table 1): three in English and three in Spanish. Facilitators followed an interview guide that included discussion about attitudinal, infrastructural, and cultural factors impacting decisions around HPV vaccination. Following this, groups provided feedback on the HPV component of *Teen Vaxscene* with regard to its overall appeal, design, layout, functionality, and content. Groups were also asked to suggest additional information and features to include in the intervention. All focus groups were audio-recorded and transcribed verbatim. Spanish transcripts were translated into English by a native Spanish speaker and validated by a second native Spanish speaker for accuracy and clarity.

Analysis

Two members of the research team reviewed CAC meeting notes in order to summarize member feedback into two main categories, which included content and design of the intervention. CAC members reviewed the summaries to confirm the meaning, clarity, and completeness of the group feedback. Focus group data analysis was conducted using established qualitative content analysis methods and reflexive team analysis (Graneheim & Lundman, 2004; Hsieh & Shannon, 2005). Transcripts and CAC meeting notes were independently read multiple times by members of the study team in order to achieve immersion. Responses were then categorized independently using an emergent, rather than *a priori*, approach into codes pertaining to factors impacting decision-making around HPV vaccination and to suggestions for the intervention. Initial code categories were compared and discussed with a third member of the study team until code agreement was achieved. Two analysts then applied the consensus codes to the transcripts, and a third confirmed the coding. The analysts met routinely to check new findings, discuss emergent new codes and themes, confirm intercoder reliability, and assess preliminary results of the analysis process. Discrepancies were resolved by mutual agreement between reviewers. Three main content codes and three subcodes emerged; multiple codes were assigned to quotes as needed due to the complexity of their content. Once all data were coded, quotes derived from different groups were considered separately to identify any difference in responses between parent versus young adult participants, or between those who participated in English- versus Spanish-facilitated groups. Throughout the analytic process, the qualitative data software program ATLAS.ti (v.7.0; Scientific Software Development GmbH, Berlin, Germany) was used for data organization and management. Feedback from the CAC and focus groups was synthesized and summarized into a list of edits for the intervention development team and incorporated into the final version of the intervention.

Results

Community Advisory Committee

Target Population for the Intervention

CAC members provided discussion-based feedback about the content and design of the website, and about ways to modify *Teen Vaxscene*. Although our initial intention was to design an intervention focused specifically on *female* HPV vaccination, it became apparent at the first CAC meeting that intervention materials should include information on male HPV vaccination. Thus, a decision was made early in the process to modify *Teen VaxScene* (created prior to recommendations for universal male HPV vaccination) to appeal to both men and women. After initial assessment of *Teen VaxScene*, members helped select the color scheme, design the logo, and make iterative changes to the content for the new intervention. Members also helped devise a name for the revised intervention (*CHICOS; Combatting HPV Infection and CancerS*), which was specifically chosen to reflect the Latino community and also be inclusive of both males and females. (The alternative, "*CHICAS*," would have suggested an exclusively female orientation.)

Design and Content

The CAC agreed that the intervention would be more appealing if the overall design appeared less "clinical," incorporated more vibrant colors, and used a more eye-catching logo. Members also noted that pages were too "wordy," and suggested it would be more effective if participants could choose which content to read about in more detail rather than being presented with everything at once. Members requested graphic photos depicting HPV-related cancers and diseases, but suggested that these be optional for viewing, and labeled such that participants could select "the body parts" (e.g., male versus female genitalia) they specifically wanted to see. Finally, the CAC indicated that an emphasis on basic information about HPV should be presented prior to any tailored content in order to make the intervention most useful.

Parent and Young Adult Focus Groups

Of the 27 parents who participated in the focus groups, three were male and 24 were female. All young adult focus group

participants were female. In all focus groups, participants were asked open-ended questions regarding their knowledge, beliefs, and feelings about the HPV vaccine. As described below, three main themes emerged from the focus groups, including barriers to vaccination, the need for additional information about HPV, and intervention modifications.

Barriers to Vaccination

Several barriers to vaccination were identified that were not specifically addressed in *Teen VaxScene* and needed to be incorporated into the modified intervention. For example, similar to the CAC, focus group participants perceived lacking enough general information about HPV and the HPV vaccine to make an informed decision about whether to get vaccinated, and suggested incorporating this information early in the interventional website. The vaccine cost was believed to be prohibitive for participants as illustrated below, suggesting that the information addressing this was critical:

> And some people are like ok, where can I go? Or how much is it going to cost me? And do I have to pay? Can I get it free? So those were the main things that I know actually affect the difference [of whether or not people will get the vaccine].

Most parents suggested that talking about the vaccine with their adolescent children would be a good conversation-starter about safe sexual practices; however, they believed that many parents would fear that teens would see the vaccine as a clearance for sexual promiscuity. As one participant articulated:

> I think people are afraid; what I've heard from other friends who have daughters, is they are afraid it is a 'free for all'. Because their daughter has [the vaccine], they think they are free and clear and they are going to be protected and they are just having sex.

Related to this, several parents suggested that the intervention would be more appropriately targeted to parents specifically, rather than parents and adolescents.

Particular to Spanish speakers were concerns of reproductive consequences associated with the HPV vaccine. Spanish speakers also asserted that preventive medicine (such as vaccinations) was not a strong cultural norm and that their families preferred home remedies. In particular, participants disclosed that if they feel "healthy" and don't have a family history of illness, they rarely schedule provider visits, as described by the following comment:

> Because also as Hispanics we think that we won't get the diseases. [You say] no, well I'm healthy, and I feel fine, my kids feel fine, and this and that, and so we refuse to go to the doctor, because we want to do everything at home, with home remedies only.

Spanish speakers also suggested that many within their community hold strong religious values around sexual abstinence and would be reluctant to get the HPV vaccine, particularly for their adolescents, as one participant disclosed:

> I think all the women in my family are very supportive of [the HPV vaccine]. But the males in my family, who have daughters, they are really kind of wary about it and in denial; 'she doesn't need it', and that kind of thing. Going back to, they don't want to encourage their teenager to have sex.

Moreover, Spanish speakers relayed that when they do visit their provider, there is little, if any, discussion around HPV or the HPV vaccine, thus contributing to their perceived need for even basic information. As one participant illustrated:

> It's for the same reason that you are telling us, really for the same reason, that the information that we get from the pediatricians is scarce, it is how we live, because they never give us an explanation really. They just say, '[your child] won't get a temperature, give them this, this, and this', but they never explain to us anything.

The Need for Additional Information

The theme of desiring additional information was pervasive among all focus group discussions. Most participants reported that while they had heard of the HPV vaccine and had mostly positive attitudes about it, because it was a relatively new vaccine they remained skeptical about its safety and effectiveness. Additionally, some participants articulated inaccurate beliefs about HPV itself (e.g., its cause, transmission, etc.), as illustrated below:

> What I have heard is that the boy is the transmitter [of HPV]. I mean what I have read is that the boy is only the transmitter of the virus.
>
> I heard that this virus is carried by large mosquitos, right?

Participants requested more introductory information about HPV on the intervention website, along with additional logistical information about the HPV vaccine. Specifically, both parents and young adults indicated that they were interested in learning about how HPV is transmitted, risk of contracting HPV, health or disease risks related to contracting HPV, symptoms of HPV, possibilities for treatment, and the extent to which males are at risk of HPV:

> The symptoms … for example, if I get the [HPV] virus, I won't have the symptoms tomorrow, or the day after tomorrow, like with time, until you do a test? So then you say, ok, you have contracted this virus, but I mean what are the symptoms to notice; temperature, fatigue, or what are they?
>
> Why isn't there a study that says that it's proven that [the HPV vaccine] protects? Better yet, I think that those of us who are not sure if it is unsafe are unsure because of this lack of information.

Most participants also agreed that they wanted to learn more about research efforts and results pertaining to the HPV vaccine, the optimal age range for getting the vaccine, when to initiate the vaccine series, and when to pursue subsequent doses. Parents in particular expressed the desire for evidence about vaccine effectiveness, side effects, and how the vaccine works with the immune system. Parent participants in groups conducted in Spanish requested more information about HPV detection and prevention along with resources related to attenuating the cost of the HPV vaccine. For many of these information needs, participants requested access to statistical

information (e.g., cancer prevalence) that is specific to the Latino population, as articulated below:

> It is important to know where and who this type of thing affects, and if it affects everybody all over the country. So to have statistics to weigh against your statistics—your own—would be a good idea.

Intervention Modifications

Participants' evaluation of *Teen VaxScene* resulted in three subcategories of modifications that would make an HPV-specific educational intervention more useful for Latinos. The first related to streamlining *content*. Overall, participants found the existing information about HPV in *Teen VaxScene* credible and useful, but stressed the need to add concise, understandable content about the HPV vaccine and about HPV itself, as described across all themes. Both parents and young adults suggested that text should be minimized and that more photos of HPV-related cancers and other diseases be included. Spanish-speaking parents reiterated the suggestion of our CAC that the photos be optional for viewing and that they contain a "warning" statement about the photos' graphic nature.

The second subcategory pertained to the intervention's *look and feel*. Like the CAC, most participants described the existing theme as too "clinical" and suggested that the design incorporates brighter colors and a more family-oriented logo. In line with minimizing the amount of text, both parents and young adults requested that more of the content be presented as graphical representations.

The third subcategory incorporated suggestions about *technical aspects* of the intervention. Participants expressed the importance of having control over what they wanted to see and read. Both parents and young adults thought users should be able to choose which disease-related photos they wanted to view, and be more easily able to select content they wanted to read about in more detail. Finally, participants suggested adding navigational features throughout the intervention such as arrows and pop-up menus describing how to navigate through the site.

Incorporation of Feedback to Create CHICOS

Figure 1 depicts screenshots of *Teen VaxScene* and the finalized intervention, *CHICOS*, after incorporating feedback from the CAC and focus groups. Members of the focus groups, especially those facilitated in Spanish, emphasized that they feel they receive little or no information about HPV or the HPV vaccine during visits with their (or their child's) primary care provider. Additionally, both the CAC and focus group members indicated that they know even less about the impact of HPV or necessity of the HPV vaccine in relation to men. A change related to this feedback was the prioritization of general information before individually tailored information, including information related to male HPV. Specifically, male-oriented material was added to the intervention to clarify the risks of HPV to men (types and frequencies of cancers), how HPV is detected in women versus men, statistics about the degree to which the HPV vaccine has been studied and administered among men, and whether vaccine side effects differ between women and men. Gender-specific HPV- and disease-related photos were also included to reflect the most common symptoms and diseases affecting both women and men.

Focus group participants and CAC members agreed that members of the Hispanic community often do not value or make use of preventive care, especially if no symptoms of illness are present. We thus added content to emphasize that HPV is most often asymptomatic, that receiving the vaccine prevents most cases of genital warts and HPV-related cancers, and that receiving the vaccine prior to beginning any sexual activity is the most effective route to prevention. The latter point was also included to address uncertainties noted among focus group members related to initiating the vaccine in adolescents who are not yet sexually active.

Despite the apparent need to add a fair amount of content to the revised intervention, both CAC and focus group participants agreed that in order to be effective and user-friendly, the webpages should not be too "wordy." We thus used an "accordion-style" format for text to minimize the amount of written text present on any given screen. This format incorporates short phrases that users can read quickly and click on, if interested, to receive expanded information and descriptions about a topic, and accomplishes an organized and simplified appearance on each webpage. Graphical representations of reported HPV-related statistics and about how the HPV vaccine works with the immune system were also included in order to decrease text and streamline information.

In terms of modifications to create a less "clinical" design, we eliminated animated depictions of virus particles incorporated into the theme of the original intervention. Additionally, our CAC chose a brighter, more pleasing color scheme and helped develop a logo that incorporated a style appropriate to a broad end-user base. Navigational changes included additional navigational instructions (e.g., "Tap here to go to a different page"; "Are you a visual learner? Tap here to see a photo of how vaccines work").

Discussion

In this study, we used iterative feedback from a Community Advisory Committee, combined with data from focus groups, to explore the design-, technical-, and content-related modifications needed to effectively adapt an existing educational intervention about HPV vaccines, called *Teen VaxScene*, into a related intervention specific for the Latino community. Several themes emerged from this process, including a need for basic information about the HPV vaccine prior to more individualized information, changes to the look and feel of the intervention to make it appear less clinical, and the incorporation of information that addressed specific barriers identified by the Latino community as important that were not included in the original intervention. This feedback has helped us to modify the prior intervention into its current form, called *CHICOS*—a name chosen by our CAC to reflect the Latino population and the change of including information for both men and women.

Consistent with prior research (Bednarczyk, Birkhead, Morse, Doleyres, & McNutt, 2011; Conroy et al., 2009; DiClemente et al., 2015; Dillard & Spear, 2010; Downs, Scarinci, Einstein, Collins, & Flowers, 2010; Ferrer, Trotter,



Fig. 1. Screenshots depicting website changes (1—Premodification, 2—Postmodification): (A) Brighter color scheme, logo, additional navigation; (B) Introductory information about HPV; (C) Choice of disease-related photos.

Hickman, & Audrey, 2014; Firenze et al., 2015; Kepka, Ding, Bodson, Warner, & Mooney, 2015; Kepka, Warner, Kinney, Spigarelli, & Mooney, 2015; Short et al., 2010), the CAC and focus groups all reported vaccine cost, access to insurance, and a general lack of awareness and/or understanding of either HPV or the vaccine as barriers to vaccination, prompting us to prioritize the presentation of this information in *CHICOS*. Somewhat surprising was *lack* of concern about stigma associated with HPV infection or vaccination, or a need to highly consider friends' and families' acceptance of the HPV vaccine as barriers to vaccination, both which have been documented in other studies about HPV vaccination within this population (Bair, Mays, Sturm, & Zimet, 2008; Brewer & Fazekas, 2007; Hopfer & Clippard, 2011; Scarinci, Garces-Palacio, & Partridge, 2007; Stephens & Thomas, 2014; Thomas, Stephens, Johnson-Mallard, & Higgins, 2014; Vanslyke et al., 2008). An important finding was that Spanish speakers in particular indicated that cultural or religious values and expectations could act as barriers to vaccination and requested information in CHICOS to address these issues specifically. Also important was the finding that all groups perceived a need for very "general" information about the vaccine prior to more nuanced or personalized information. This finding is in keeping with other studies demonstrating that a limited understanding of the risks and benefits of the vaccine was a major reason for not starting or completing the vaccine series among Latinos (as well as other populations) (Fernandez et al., 2009; Kepka, Warner et al., 2015; Luque, Raychowdhury, & Weaver, 2012; O'Leary, Barnard, Lockhart, Albright, Jimenez-Zambrano, Lee, Stokley, Dempsey, Kempe, 2014; Vanslyke et al., 2008; Warner et al., 2015).

In terms of navigation, the educational intervention prototype generated suggestions from all groups to reduce the amount of text such that the information was clear and concise, to use brighter and more appealing colors, to create an eye-catching logo, to provide enough user direction to make navigation easy and clear, and to increase the amount of navigational control available to the user. Based on participant

feedback, we also presented graphics depicting how the HPV vaccine works with the immune system and text about research efforts and risk and protective results pertaining to the HPV vaccine.

Although positive outcomes across a wide range of clinical outcomes have been indicated (Davies, Spence, Vandelanotte, Caperchione, & Mummery, 2012; Lustria et al., 2013; Nguyen, Carrieri-Kohlman, Rankin, Slaughter, & Stulbarg, 2004), very little formal evidence exists concerning which components of web-based interventions make them more effective (Dempsey et al., 2015; Dempsey & Zimet, 2015; Fu, Bonhomme, Cooper, Joseph, & Zimet, 2014; Starling et al., 2014; Zhang, Tsark, Campo, & Teti, 2015). Among studies using interventions to increase adolescent HPV vaccination, results have been limited by various factors such as small sample size, a limited age group, and measuring only intention to vaccinate rather than actual vaccine uptake (Dempsey et al., 2011; Dempsey et al., 2015; DiClemente et al., 2015; Fu et al., 2014). Additionally, few studies have included community members in the development of web-based vaccine-related interventions even though a community-based participatory approach to the design of interventions has been demonstrated to enhance their overall effectiveness (Katz & Paskett, 2015). Still fewer studies have tested the effect of cultural tailoring for vaccine-related interventions (DiClemente et al., 2015). For these reasons, we undertook this work in preparation for a formal evaluation of these factors in a randomized intervention trial among multiple integrated clinics to compare the use and effectiveness of the intervention's interactive technology with usual care. If effective, our web-based educational intervention could represent a useful resource to reduce HPV vaccination concerns, improve immunization rates, and educate Latino young adults and parents of adolescents about the HPV vaccine and HPV itself outside of the time boundaries of the traditional clinical encounter.

This study has several limitations. First, the CAC and focus group were comprised of people who chose to be involved in these discussions. These participants may therefore be more involved in community issues, have higher health literacy, or have stronger opinions about the HPV vaccine than average end users for the intervention. Second, there is the possibility of social desirability bias causing participants to provide more desirable feedback about the website when faced with the research team. Mitigating this somewhat is the consistency of our focus group data with that of others regarding barriers to HPV vaccination and the fact that we have had an additional two years of ongoing feedback from the CAC after these data were collected and integrated into the website without substantial changes in their opinions about the current CHICOS intervention. Third, very few study participants were men. Although three fathers of adolescents participated in the focus groups, none of the young adult participants or CAC members were male, which leaves some gaps in the knowledge of what information and format male end users of the intervention may find most useful or appealing. Moreover, the study results reflect discussion-based data and did not include any direct observations of patients using the intervention components, which may uncover issues that influence its usefulness. Finally, we do not yet know whether the developed intervention is effective in improving HPV vaccination rates. The impact of this newly developed web-based intervention is currently being evaluated in a randomized clinical trial.

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References

Bair, R. M., Mays, R. M., Sturm, L. A., & Zimet, G. D. (2008). Acceptability of the human papillomavirus vaccine among Latina mothers. *Journal of Pediatric and Adolescent Gynecology*, *21* (6), 329–334. doi:10.1016/j.jpag.2008.02.007

Bednarczyk, R. A., Birkhead, G. S., Morse, D. L., Doleyres, H., & McNutt, L. A. (2011). Human papillomavirus vaccine uptake and barriers: Association with perceived risk, actual risk and race/ethnicity among female students at a New York State university, 2010. *Vaccine*, *29* (17), 3138–3143. doi:10.1016/j.vaccine.2011.02.045

Brewer, N. T., & Fazekas, K. I. (2007). Predictors of HPV vaccine acceptability: A theory-informed, systematic review. *Preventive Medicine*, *45* (2–3), 107–114. doi:10.1016/j.ypmed.2007.05.013

Centers for Disease Control and Prevention. (2004). *Genital HPV Infection - CDC Fact Sheet*. Retrieved from http://www.cdc.gov/std/HPV/STDFact-HPV.htm

Centers for Disease Control and Prevention. (2012, April 20). Human papillomavirus-associated cancers - United States, 2004–2008. *MMWR Morb Mortal Wkly Rep*. 2012/04/20. Retrieved from http://www.cdc.gov/mmwr/preview/mmwrhtml/mm6115a2.htm

Cokkinides, V. E., Bandi, P., Siegel, R. L., & Jemal, A. (2012). Cancer-related risk factors and preventive measures in US Hispanics/Latinos. *CA: A Cancer Journal for Clinicians*, *62* (6), 353–363. doi:10.3322/caac.21155

Conroy, K., Rosenthal, S. L., Zimet, G. D., Jin, Y., Bernstein, D. I., Glynn, S., & Kahn, J. A. (2009). Human Papillomavirus Vaccine Uptake, Predictors of Vaccination, and Self-Reported Barriers to Vaccination. *Journal of Women's Health*, *18* (10), 1679–1686. doi:10.1089/jwh.2008.1329

Davies, C. A., Spence, J. C., Vandelanotte, C., Caperchione, C. M., & Mummery, W. K. (2012). Meta-analysis of internet-delivered interventions to increase physical activity levels. *International Journal of Behavioral Nutrition and Physical Activity*, *9*, 13. doi:10.1186/1479-5868-9-52

Davis, T., Fredrickson, D., Arnold, C., Cross, J., Humiston, S., Green, K., & Bocchini, J. J. (2001). Childhood vaccine risk/benefit communication in private practice office settings: A national survey. *Pediatrics*, *107*(2), E17. doi:10.1542/peds.107.2.e17

Dempsey, A. F., Maertens, J., Beaty, B. L., & O'Leary, S.T. (2015). Understanding how different recruitment strategies impact parent engagement with an iPad-based intervention to provide personalized information about adolescent vaccines. *Journal of Adolescent Health*, *55* (5 Suppl), S7–13.

Dempsey, A. F., Schaffer, S., Barr, K., Ruffin, M. T., & Carlos, R. (2011). Improving maternal intention for HPV vaccination using tailored

educational materials. In: *Abstract Book 1, 27th International Papillomavirus Society Meeting, Berlin, Germany* (Abstract P-01.19, p. 15).

Dempsey, A. F., & Zimet, G. D. (2015). Interventions to Improve Adolescent Vaccination: What May Work and What Still Needs to Be Tested. *American Journal of Preventive Medicine*, *49*, S445–S454. doi:10.1016/j.amepre.2015.04.013

DiClemente, R. J., Murray, C. C., Graham, T., & Still, J. (2015). Overcoming barriers to HPV vaccination: A randomized clinical trial of a culturally-tailored, media intervention among African American girls. *Human Vaccines and Immunotherapeutics, 11*, 2883–2894. doi:10.1080/21645515.2015.1070996

Dillard, J. P., & Spear, M. E. (2010). Knowledge of human papillomavirus and perceived barriers to vaccination in a sample of US female college students. *Journal of American College Health*, *59* (3), 186–190. doi:10.1080/07448481.2010.493189

Downs, L. S., Jr., Scarinci, I., Einstein, M. H., Collins, Y., & Flowers, L. (2010). Overcoming the barriers to HPV vaccination in high-risk populations in the US. *Gynecologic Oncology*, *117* (3), 486–490. doi:10.1016/j.ygyno.2010.02.011

Fernandez, M. E., McCurdy, S. A., Arvey, S. R., Tyson, S. K., Morales-Campos, D., Flores, B., … Sanderson, M. (2009). HPV knowledge, attitudes, and cultural beliefs among Hispanic men and women living on the Texas-Mexico border. *Ethnicity & Health*, *14* (6), 607–624. doi:10.1080/13557850903248621

Ferrer, H. B., Trotter, C., Hickman, M., & Audrey, S. (2014). Barriers and facilitators to HPV vaccination of young women in high-income countries: A qualitative systematic review and evidence synthesis. *BMC Public Health*, *14*, 700. doi:10.1186/1471-2458-14-700

Firenze, A., Marsala, M. G., Bonanno, V., Maranto, M., Ferrara, C., Giovannelli, L., & Restivo, V. (2015). Facilitators and barriers HPV unvaccinated girls after 5 years of program implementation. *Human Vaccines & Immunotherapeutics*, *11* (1), 240–244. doi:10.4161/hv.36158

Fu, L. Y., Bonhomme, L. A., Cooper, S. C., Joseph, J. G., & Zimet, G. D. (2014). Educational interventions to increase HPV vaccination acceptance: A systematic review. *Vaccine*, *32* (17), 1901–1920. doi:10.1016/j.vaccine.2014.01.091

Gargano, J. W., Wilkinson, E. J., Unger, E. R., Steinau, M., Watson, M., Huang, Y. J., … Saraiya, M. (2012). Prevalence of Human Papillomavirus Types in Invasive Vulvar Cancers and Vulvar Intraepithelial Neoplasia 3 in the United States Before Vaccine Introduction. *Journal of Lower Genital Tract Disease*, *16* (4), 471–479. doi:10.1097/LGT.0b013e3182472947

Gerend, M. A., Shepherd, M. A., & Lustria, M. L. (2013). Increasing human papillomavirus vaccine acceptability by tailoring messages to young adult women's perceived barriers. *Sexually Transmitted Diseases*, *40* (5), 401–405. doi:10.1097/OLQ.0b013e318283c8a8

Glasgow, R. E., Bull, S. S., Piette, J. D., & Steiner, J. F. (2004). Interactive behavior change technology. A partial solution to the competing demands of primary care. *American Journal of Preventive Medicine*, *27* (2), 80–87. doi:10.1016/j.amepre.2004.04.026

Glenn, B., Tsui, J., Coronado, G., Fernandez, M., Savas, L., Taylor, V., & Bastani, R. (2015). Understanding HPV Vaccination Among Latino Adolescent Girls in Three U.S. Regions. *Journal of Immigrant and Minority Health*, *17* (1), 96–103. doi:10.1007/s10903-014-9996-8

Gowda, C., Schaffer, S. E., Dombkowski, K. J., & Dempsey, A. F. (2012a). Understanding attitudes toward adolescent vaccination and the decision-making dynamic among adolescents, parents and providers. *BMC Public Health*, *12*, 509. doi:10.1186/1471-2458-12-509

Gowda, C., Schaffer, S. E., Dombkowski, K. J., & Dempsey, A. F. (2012b). Understanding attitudes toward adolescent vaccination and the decision-making dynamic among adolescents, parents and providers. *BMC Public Health*, *12* (1), 509. doi:10.1186/1471-2458-12-509

Gowda, C., Schaffer, S. E., Kopec, K., Markel, A., & Dempsey, A. F. (2012). A pilot study on the effects of individually-tailored education for MMR vaccine-hesitant parents on MMR vaccination intention. *Human Vaccines & Immunotherapeutics*, *9*, 437–445.

Graneheim, U. H., & Lundman, B. (2004). Qualitative content analysis in nursing research: Concepts, procedures and measures to achieve trustworthiness. *Nurse Education Today*, *24* (2), 105–112. doi:10.1016/j.nedt.2003.10.001

Hopenhayn, C., Christian, A., Christian, W. J., Watson, M., Unger, E. R., Lynch, C. F., … Saraiya, M. (2014). Prevalence of Human Papillomavirus Types in Invasive Cervical Cancers From 7 US Cancer Registries Before Vaccine. *Journal of Lower Genital Tract Disease*, *18* (2), 182–189. doi:10.1097/LGT.0b013e3182a577c7

Hopfer, S., & Clippard, J. R. (2011). College women's HPV vaccine decision narratives. *Qualitative Health Research*, *21* (2), 262–277. doi:10.1177/1049732310383868

Hsieh, H. F., & Shannon, S. E. (2005). Three approaches to qualitative content analysis. *Qualitative Health Research*, *15* (9), 1277–1288. doi:10.1177/1049732305276687

Jimbo, M., Nease, D. E., Jr., Ruffin, M. T. T., & Rana, G. K. (2006). Information technology and cancer prevention. *CA: A Cancer Journal for Clinicians*, *56* (1), 26–36; quiz 48-29. doi:10.3322/canjclin.56.1.26

Katz, M. L., & Paskett, E. D. (2015). The process of engaging members from two underserved populations in the development of interventions to promote the uptake of the HPV vaccine. *Health Promotion Practice*, *16* (3), 443–453. doi:10.1177/1524839914559776

Kempe, A., Daley, M. F., McCauley, M. M., Crane, L. A., Suh, C. A., Kennedy, A. M., … Dickinson, L. M. (2011). Prevalence of parental concerns about childhood vaccines: The experience of primary care physicians. *American Journal of Preventive Medicine*, *40* (5), 548–555. doi:10.1016/j.amepre.2010.12.025

Kepka, D., Ding, Q., Bodson, J., Warner, E. L., & Mooney, K. (2015). Latino Parents' Awareness and Receipt of the HPV Vaccine for Sons and Daughters in a State with Low Three-Dose Completion. *Journal of Cancer Education*, *30*, 808–812. doi:10.1007/s13187-014-0781-0

Kepka, D., Warner, E. L., Kinney, A. Y., Spigarelli, M. G., & Mooney, K. (2015). Low Human Papillomavirus (HPV) Vaccine Knowledge Among Latino Parents in Utah. *Journal of Immigrant and Minority Health*, *17* (1), 125–131. doi:10.1007/s10903-014-0003-1

Kreuter, M. W., Caburnay, C. A., Chen, J. J., & Donlin, M. J. (2004). Effectiveness of individually tailored calendars in promoting childhood immunization in urban public health centers. *American Journal of Public Health*, *94* (1), 122–127. doi:10.2105/AJPH.94.1.122

Kreuter, M. W., Strecher, V. J., & Glassman, B. (1999). One size does not fit all: The case for tailoring print materials. *Ann Behav Med*, *21* (4), 276–283. doi:10.1007/BF02895958

Kreuter, M. W., & Wray, R. J. (2003). Tailored and targeted health communication: Strategies for enhancing information relevance. *American Journal of Health Behavior*, *27* (Suppl 3), S227–S232. doi:10.5993/AJHB.27.1.s3.6

Laugesen, J., Hassanein, K., & Yuan, Y. (2015). The Impact of Internet Health Information on Patient Compliance: A Research Model and an Empirical Study. *Journal of Medical Internet Research*, *17* (6), e143. doi:10.2196/jmir.4333

Ludden, G. D., van Rompay, T. J., Kelders, S. M., & van Gemert-Pijnen, J. E. (2015). How to Increase Reach and Adherence of Web-Based Interventions: A Design Research Viewpoint. *Journal of Medical Internet Research*, *17* (7), e172. doi:10.2196/jmir.4201

Luque, J. S., Raychowdhury, S., & Weaver, M. (2012). Health care provider challenges for reaching Hispanic immigrants with HPV vaccination in rural Georgia. *Rural Remote Health*, *12* (2), 1975.

Lustria, M. L., Noar, S. M., Cortese, J., Van Stee, S. K., Glueckauf, R. L., & Lee, J. (2013). A meta-analysis of web-delivered tailored health behavior change interventions. *Journal of Health Communication*, *18* (9), 1039–1069. doi:10.1080/10810730.2013.768727

Markowitz, L. E., Dunne, E. F., Saraiya, M., Chesson, H. W., Curtis, C. R., Gee, J., … Unger, E. R. (2014). Human papillomavirus vaccination:

Recommendations of the Advisory Committee on Immunization Practices (ACIP). *MMWR Recomm Rep*, *63* (RR–05), 1–30.

Morrison, L., Moss-Morris, R., Michie, S., & Yardley, L. (2014). Optimizing engagement with Internet-based health behaviour change interventions: Comparison of self-assessment with and without tailored feedback using a mixed methods approach. *British Journal of Health Psychology*, *19* (4), 839–855. doi:10.1111/bjhp.12083

National Cancer Institute. *Cancer health disparities fact sheet*. Retrieved from http://www.cancer.gov/cancertopics/factsheet/disparities/cancer-health-disparities

Nguyen, H. Q., Carrieri-Kohlman, V., Rankin, S. H., Slaughter, R., & Stulbarg, M. S. (2004). Internet-based patient education and support interventions: A review of evaluation studies and directions for future research. *Computers in Biology and Medicine*, *34* (2), 95–112. doi:10.1016/s0010-4825(03)00046-5

O'Leary, S. T., Crane, L. A., Wortley, P., Daley, M. F., Hurley, L. P., Dong, F., … Kempe, A. (2012). Adherence to Expanded Influenza Immunization Recommendations among Primary Care Providers. *The Journal of Pediatrics*, *160* (3), 480–486 e481. doi:10.1016/j.jpeds.2011.09.010

Pierre Joseph, N., Belizaire, M., Porter, C. L., Walsh, J. P., Esang, M., Goff, G., & Perkins, R. B. (2014). Ethnic Differences in Perceived Benefits and Barriers to HPV Vaccine Acceptance: A Qualitative Analysis of Young African American, Haitian, Caucasian, and Latino Men. *Clinical Pediatrics*, *53* (2), 177–185. doi:10.1177/0009922813515944

Podolsky, R., Cremer, M., Atrio, J., Hochman, T., & Arslan, A. A. (2009). HPV vaccine acceptability by Latino parents: A comparison of U.S. and Salvadoran populations. *Journal of Pediatric and Adolescent Gynecology*, *22* (4), 205–215. doi:10.1016/j.jpag.2008.05.010

Rambout, L., Tashkandi, M., Hopkins, L., & Tricco, A. C. (2014). Self-reported barriers and facilitators to preventive human papillomavirus vaccination among adolescent girls and young women: A systematic review. *Preventive Medicine*, *58*, 22–32. doi:10.1016/j.ypmed.2013.10.009

Reagan-Steiner, S., Yankey, D., Jeyarajah, J., Elam-Evans, L. D., Singleton, J. A., Curtis, C. R., … Stokley, S. (2015). National, Regional, State, and Selected Local Area Vaccination Coverage Among Adolescents Aged 13-17 Years - United States, 2014. *MMWR Morb Mortal Wkly Rep*, *64* (29), 784–792. doi:10.15585/mmwr.mm6429a3

Rowsell, A., Muller, I., Murray, E., Little, P., Byrne, C. D., Ganahl, K., … Yardley, L. (2015). Views of People With High and Low Levels of Health Literacy About a Digital Intervention to Promote Physical Activity for Diabetes: A Qualitative Study in Five Countries. *Journal of Medical Internet Research*, *17* (10), 12. doi:10.2196/jmir.4999

Scarinci, I. C., Garces-Palacio, I. C., & Partridge, E. E. (2007). An Examination of Acceptability of HPV Vaccination among African American Women and Latina Immigrants. *Journal of Women's Health*, *16* (8), 1224–1233. doi:10.1089/jwh.2006.0175

Schneider, F., Van Osch, L., & De Vries, H. (2012). Identifying Factors for Optimal Development of Health-Related Websites: A Delphi Study Among Experts and Potential Future Users. *Journal of Medical Internet Research*, *14* (1), e18. doi:10.2196/jmir.1863

SEER. StatFacts: Cervical Cancer. Retrieved from http://seer.cancer.gov/statfacts/html/cervix.html

Short, M. B., Rosenthal, S. L., Sturm, L., Black, L., Loza, M., Breitkopf, D., & Zimet, G. D. (2010). Adult women's attitudes toward the HPV vaccine. *Journal of Women's Health*, *19* (7), 1305–1311. doi:10.1089/jwh.2009.1471

Shoup, J. A., Wagner, N. M., Kraus, C. R., Narwaney, K. J., Goddard, K. S., & Glanz, J. M. (2015). Development of an interactive social media tool for parents with concerns about vaccines. *Health Education & Behavior*, *42* (3), 302–312. doi:10.1177/1090198114557129

Siegel, R. L., Fedewa, S. A., Miller, K. D., Goding-Sauer, A., Pinheiro, P. S., Martinez-Tyson, D., & Jemal, A. (2015). Cancer statistics for Hispanics/Latinos, 2015. *CA: A Cancer Journal for Clinicians*, *65*, 457–480. doi:10.3322/caac.21314

Springvloet, L., Lechner, L., De Vries, H., Candel, M. J., & Oenema, A. (2015). Short- and medium-term efficacy of a web-based computer-tailored nutrition education intervention for adults including cognitive and environmental feedback: Randomized controlled trial. *Journal of Medical Internet Research*, *17* (1), e23. doi:10.2196/jmir.3837

Starling, R., Nodulman, J. A., Kong, A. S., Wheeler, C. M., Buller, D. B., & Woodall, W. G. (2014). Beta-test Results for an HPV Information Web site: GoHealthyGirls.org - Increasing HPV Vaccine Uptake in the United States. *Journal of Consumer Health on the Internet*, *18* (3), 226–237. doi:10.1080/15398285.2014.931771

Steinau, M., Unger, E. R., Hernandez, B. Y., Goodman, M. T., Copeland, G., Hopenhayn, C., … Saraiya, M. (2013). Human papillomavirus prevalence in invasive anal cancers in the United States before vaccine introduction. *Journal of Lower Genital Tract Disease*, *17* (4), 397–403. doi:10.1097/LGT.0b013e31827ed372

Stephens, D. P., & Thomas, T. L. (2014). Social Networks Influence Hispanic College Women's HPV Vaccine Uptake Decision-making Processes. *Women's Reproductive Health*, *1* (2), 120–137. doi:10.1080/23293691.2014.966034

Thomas, T. L., Stephens, D. P., Johnson-Mallard, V., & Higgins, M. (2014). Young Hispanic Men and Human Papillomavirus Vaccination Choices. *Journal of Transcultural Nursing: Official Journal of the Transcultural Nursing Society / Transcultural Nursing Society*. doi:10.1177/1043659614526759

U.S. Department of Health and Human Services. (2010). *Healthy People 2020*. Retrieved from www.healthypeople.gov

Ubhi, H. K., Michie, S., Kotz, D., Wong, W. C., & West, R. (2015). A Mobile App to Aid Smoking Cessation: Preliminary Evaluation of SmokeFree28. *Journal of Medical Internet Research*, *17* (1), e17. doi:10.2196/jmir.3479

Vanslyke, J. G., Baum, J., Plaza, V., Otero, M., Wheeler, C., & Helitzer, D. L. (2008). HPV and cervical cancer testing and prevention: Knowledge, beliefs, and attitudes among Hispanic women. *Qualitative Health Research*, *18* (5), 584–596. doi:10.1177/1049732308315734

Warner, E. L., Lai, D., Carbajal-Salisbury, S., Garza, L., Bodson, J., Mooney, K., & Kepka, D. (2015). Latino Parents' Perceptions of the HPV Vaccine for Sons and Daughters. *Journal of Community Health*, *40* (3), 387–394. doi:10.1007/s10900-014-9949-0

Watts, L. A., Joseph, N., Wallace, M., Rauh-Hain, J. A., Muzikansky, A., Growdon, W. B., & Del Carmen, M. G. (2009). HPV vaccine: A comparison of attitudes and behavioral perspectives between Latino and non-Latino women. *Gynecologic Oncology*, *112* (3), 577–582. doi:10.1016/j.ygyno.2008.12.010

Williams, W. W., Lu, P. J., O'Halloran, A., Bridges, C. B., Kim, D. K., Pilishvili, T., … Markowitz, L. E. (2015). Vaccination coverage among adults, excluding influenza vaccination - United States, 2013. *MMWR Morb Mortal Wkly Rep*, *64* (4), 95–102.

Yardley, L., Morrison, L., Bradbury, K., & Muller, I. (2015). The person-based approach to intervention development: Application to digital health-related behavior change interventions. *Journal of Medical Internet Research*, *17* (1), e30. doi:10.2196/jmir.4055

Yeganeh, N., Curtis, D., & Kuo, A. (2010). Factors influencing HPV vaccination status in a Latino population; and parental attitudes towards vaccine mandates. *Vaccine*, *28* (25), 4186–4191. doi:10.1016/j.vaccine.2010.04.010

Zhang, N., Tsark, J., Campo, S., & Teti, M. (2015). Facebook for Health Promotion: Female College Students' Perspectives on Sharing HPV Vaccine Information Through Facebook. *Hawaii J Med Public Health*, *74* (4), 136–140. Retrieved from http://www.ncbi.nlm.nih.gov/pmc/articles/PMC4407456/pdf/hjmph7404_0136.pdf